

Companion to Bird & de Moor, Algebra of Programming

§1. Programs

§1.1. Datatypes

§1.2. Natural numbers

§1.3. Lists

§1.4. Trees

§1.5. Inverses

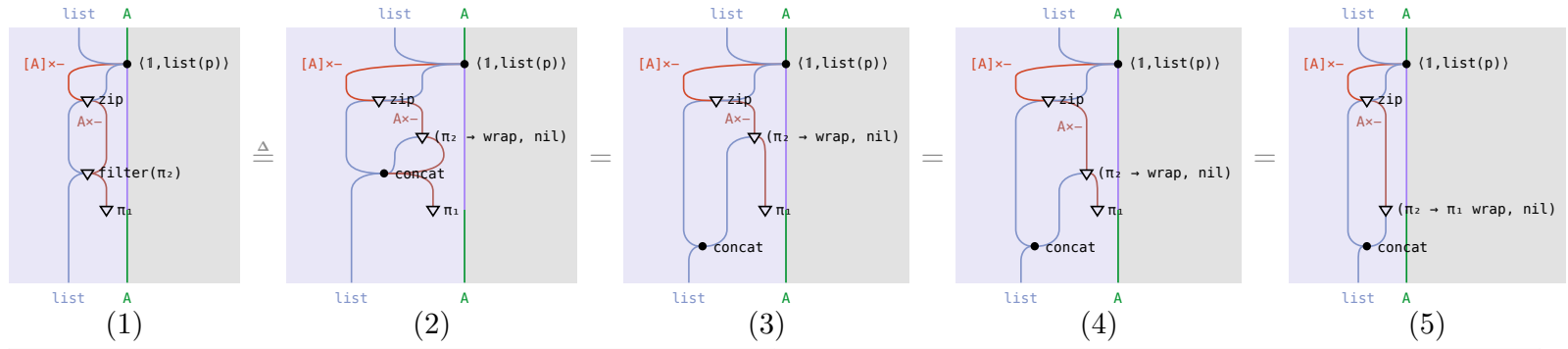
§1.6. Polymorphic functions

§1.7. Pointwise and point-free

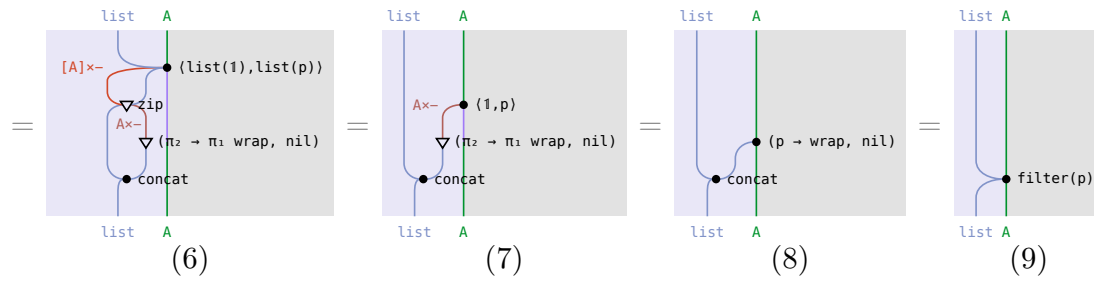
$$\langle 1, \text{list}(p) \rangle \text{ zip filter}(\pi_2) \text{ list}(\pi_1) = \text{filter}(p) \quad (1.7a)$$

pairing each element with its answer under p ($\langle 1, \text{list}(p) \rangle \text{ zip}$), keeping the pairs answered **true** ($\text{filter}(\pi_2)$) and dropping the answers ($\text{list}(\pi_1)$) is $\text{filter}(p)$

the starting composite



1	$\langle 1, \text{list}(p) \rangle \text{ zip filter}(\pi_2) \text{ list}(\pi_1)$	definition of filter: $\text{filter}(p) \triangleq \text{list}((p \rightarrow \text{wrap}, \text{nil}))$ concat
2	$\triangleq \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \text{wrap}, \text{nil})) \text{ concat list}(\pi_1)$	(1.6): $\text{concat list}(f) = \text{list}(\text{list}(f)) \text{ concat}$
3	$= \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \text{wrap}, \text{nil})) \text{ list}(\text{list}(\pi_1)) \text{ concat}$	(1.8) backwards: $\text{list}(f) \text{ list}(g) = \text{list}(fg)$
4	$= \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \text{wrap}, \text{nil}) \text{ list}(\pi_1)) \text{ concat}$	(1.11): $(p \rightarrow f, g)h = (p \rightarrow fh, gh)$; (1.5): $\text{wrap list}(f) = f \text{ wrap}$; (1.4): $\text{nil list}(f) = \text{nil}$
5	$= \langle 1, \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})) \text{ concat}$	(1.9) backwards: $1 = \text{list}(1)$



6	$= \langle \text{list}(1), \text{list}(p) \rangle \text{ zip list}((\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})) \text{ concat}$	(1.7): $\langle \text{list}(f), \text{list}(g) \rangle \text{ zip} = \text{list}(\langle f, g \rangle)$ (1.8) backwards: $\text{list}(f) \text{ list}(g) = \text{list}(fg)$
7	$= \text{list}(\langle 1, p \rangle (\pi_2 \rightarrow \pi_1 \text{ wrap}, \text{nil})) \text{ concat}$	(1.10): $h(p \rightarrow f, g) = (hp \rightarrow hf, hg)$; (1.1): $\langle f, g \rangle \pi_1 = f$; (1.3): $f \text{ nil} = \text{nil}$
8	$= \text{list}((p \rightarrow \text{wrap}, \text{nil})) \text{ concat}$	(1.12): $1 f = f$
9	$= \text{filter}(p)$	definition of filter